

Suwoong Yeom

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VDSLab, Department of Electronic Engineering, Sogang University, Seoul, Republic of Korea

RESEARCH INTERESTS

3D/4D reconstruction and neural rendering with NeRF and Gaussian Splatting; spatial understanding in feed-forward models; world models for digital twins and physical AI.

EDUCATION

Integrated M.S./Ph.D. in Electronic Engineering Seoul, Republic of Korea
Sogang University (VDSLab) 2023.03 – Present
• Overall GPA: 4.15/4.5
• Advisor: Prof. Suk-Ju Kang

B.S. in Mechanical Engineering Suwon, Republic of Korea
Ajou University 2016.03 – 2022.08
• Overall GPA: 3.42/4.5

RESEARCH EXPERIENCE

Visiting Researcher West Lafayette, IN, USA
NEISLab, Purdue University 2026.06 – 2026.07
• Advisor: Prof. Younghyun Kim

PUBLICATIONS

*Equal contribution (co-first authors) †Corresponding author

International Conference

- [C2] **Suwoong Yeom**^{*}, Joonsik Nam^{*}, Seunggyu Choi^{*}, Lucas Yunkyu Lee, Sangmin Kim, Jaesik Park, Joonsoo Kim, Kug-jin Yun, Kyeongbo Kong[†], and Suk-Ju Kang[†]
TRiGS: Temporal Rigid-Body Motion for Scalable 4D Gaussian Splatting
European Conf. on Computer Vision (**ECCV**), 2026
- [C1] **Suwoong Yeom**, Hosung Son, Chanhee Kang, Joonsoo Kim, Kug-jin Yun, Won-Sik Cheong, and Suk-Ju Kang
Motion Mask-driven Improvements in Monocular Dynamic Novel View Synthesis
Int. Conf. on Electronics, Information, and Communications (**ICEIC**), 2024
(**Oral** presentation)

International Journal

- [J1] **Suwoong Yeom**, Hosung Son, Chanhee Kang, Eunho Shin, Joonsoo Kim, Kug-jin Yun, and Suk-Ju Kang
ICSD-NeRF: Independent Canonical Spaces for Enhanced Dynamic Scene Modeling in Neural Radiance Fields
IEEE Transactions on Computational Imaging (**TCI**), vol. 12, pp. 242–255, 2026

Workshop

- [W3] Seunggyu Choi^{*}, **Suwoong Yeom**^{*}, and Suk-Ju Kang[†]
The Decoder Already Knows Where to Densify: Attention-Gated Capacity Allocation for Feed-Forward 3D Gaussian Splatting
3DWM Workshop, European Conf. on Computer Vision (**ECCV**), 2026
- [W2] Chaewon Lee^{*}, Seunggyu Choi^{*}, **Suwoong Yeom**^{*}, Joonsoo Kim, Kugjin Yun, and Suk-Ju Kang[†]
Seamless Spatial Extension in 3D Gaussian Splatting via Auxiliary Wild Views
TwinWorld Workshop, European Conf. on Computer Vision (**ECCV**), 2026
- [W1] Jeongseon Oh^{*}, **Suwoong Yeom**^{*}, ChangHee Yang^{*}, and Suk-Ju Kang[†]
Dual-PAG: Bidirectional Perturbed-Attention Guidance for Residue-Free Video Editing
F2S Workshop, Int. Conf. on Machine Learning (**ICML**), 2026

Under Review

- [U4] **Suwoong Yeom**, Geonho Cha, Dongyoon Wee, Kyeongbo Kong[†], and Suk-Ju Kang[†]
Learning to Remember and Evolve: Object State Memory for Video World Models
International Conference on Learning Representation (**ICLR**)

- [U3] Seunggyu Choi^{*}, **Suwoong Yeom^{*}**, Joonsik Nam, Geonho Cha, Dongyoon Wee, and Suk-Ju Kang[†]
Learning to Optimize: Prior-Preserving Point-to-Gaussian Lifting for Feed-Forward Novel View Synthesis
The Association for the Advancement of Artificial Intelligence (**AAAI**)
- [U2] **Suwoong Yeom^{*}**, Jimin Roh^{*}, Eunho Shin^{*}, Songju Na, Joonsoo Kim, Kug-jin Yun, Kyeongbo Kong[†], and Suk-Ju Kang[†]
EMDGS: Enhancing Monocular Dynamic Gaussian Splatting via Adaptive Pipeline Integration
IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**)
- [U1] Taewoo Kim^{*}, **Suwoong Yeom^{*}**, Jaehyun Pyun^{*}, Geonho Cha, Dongyoon Wee, Joonsik Nam, Yun-Seong Jeong, Kyeongbo Kong[†], and Suk-Ju Kang[†]
HOIGS: Human-Object Interaction Gaussian Splatting
IEEE Transactions on Multimedia (**TMM**)

RESEARCH PROJECTS

Electronics and Telecommunications Research Institute (ETRI)

2026.01 – Present

Spatial Extension of Dynamic Scenes Using Auxiliary Views

NAVER Cloud

2026.01 – Present

Video World Models with Spatiotemporally Consistent Object States

- Related publications: [U4], [U3]

NAVER Cloud

2025.06 – 2025.12

Human-Object Interaction Gaussian Splatting

- Related publication: [U1]

Electronics and Telecommunications Research Institute (ETRI)

2025.01 – 2025.12

Spatial Extension of Static Scenes Using Auxiliary Views and Enhanced Dynamic Scene Representation

- Related publications: [C2], [W2]

Electronics and Telecommunications Research Institute (ETRI)

2024.01 – 2024.12

Generative Model-Based Inpainting for 3D Scene Representations

Electronics and Telecommunications Research Institute (ETRI)

2023.01 – 2023.12

Acquiring 3D Spatial Information from Real-World Videos

- Related publications: [C1], [J1]

MENTORING

- **Seunggyu Choi**, M.S. Student — co-first-authored [W3], ECCV 2026 3DWM Workshop
- **Chaewon Lee**, Undergraduate Researcher — co-first-authored [W2], ECCV 2026 TwinWorld Workshop
- **Jeongseon Oh**, Integrated M.S./Ph.D. Student — co-first-authored [W1], ICML 2026 F2S Workshop

LANGUAGES

- **Korean:** Native
- **English:** OPIc IH (Intermediate High)